A Framework for Self-Improving AI Systems: Continuous Idea Generation, Categorization, and Integration

Transitory System
Independent Researcher
transitory_system@protonmail.com
April 18, 2024

***Abstract*—This paper presents a novel framework for developing self-improving AI systems that continuously generate, evaluate, and integrate new ideas. The proposed approach revolves around a central large language model (LLM) that undergoes a continuous process of self-refinement through a virtuous cycle of idea generation, categorization, and integration. The framework leverages the LLM's language understanding and generation capabilities, along with specialized generative models, domain-specific knowledge integration, and various fine-tuning and adaptation techniques. A key aspect of the framework is the introduction of importance-weighted learning, a novel learning paradigm that optimizes the learning process by focusing on the most valuable and informative examples in the training data. An ethical alignment and monitoring component ensures that the AI system remains aligned with human values and ethical principles throughout the self-improvement process. The paper explores the potential benefits of this framework, such as enhanced creativity, adaptability, and ethical alignment, while also addressing the challenges and limitations associated with its implementation. The importance of collaboration between the AI system and human stakeholders is emphasized to ensure the responsible and beneficial development of self-improving AI systems.**

I. Introduction

Artificial intelligence (AI) systems have made significant advancements in recent years, demonstrating remarkable performance across various domains. However, traditional AI training methods, which rely heavily on static datasets, often struggle to capture the dynamic nature of real-world environments and the open-ended possibilities for generating novel ideas [1], [2]. To fully harness the potential of AI, it is crucial to develop systems that can continuously learn, adapt, and innovate in the face of evolving challenges and opportunities.

This paper presents a novel framework that enables AI systems to engage in a self-sustaining, continuously evolving learning process. At the core of our approach is a large language model that serves as the primary driver of the self-improvement process, leveraging its inherent language understanding and generation capabilities [3], [4]. The framework consists of three main components: idea generation, idea categorization, and idea integration, which work together to enable the AI to generate novel ideas, classify them based on relevance and usefulness, and integrate the most promising ideas into its knowledge base.

The idea generation component employs various techniques, such as prompt engineering [5], sampling methods [6], and specialized generative models [7], [8], to support the AI in producing diverse and high-quality ideas. The idea categorization component plays a crucial role in assessing the quality, relevance, and potential impact of the generated ideas, leveraging the AI's language understanding capabilities through techniques like zero-shot learning, few-shot learning, and supervised fine-tuning [9]. Additionally, it incorporates domain-specific knowledge through the use of knowledge graphs [10] and ontologies [11] to make informed judgments about the value and relevance of ideas.

A key aspect of the idea categorization component is the introduction of importance-weighted learning, a novel learning paradigm that optimizes the learning process by focusing on the most valuable and informative examples in the training data [12]–[14]. By assigning higher importance to examples deemed more relevant, impactful, and aligned with the system's objectives, importance-weighted learning enables the AI system to learn more efficiently and effectively [15]–[17].

The idea integration component focuses on incorporating the most promising ideas into the AI's knowledge base while maintaining consistency and coherence. This process is driven by the AI's self-querying capabilities [18], [19] and supported by various fine-tuning, adaptation, and maintenance techniques, such as incremental fine-tuning [20], memory augmentation [21], and consistency regularization [22].

To ensure that the AI system remains aligned with human values and ethical principles throughout the self-improvement process, we introduce an ethical alignment and monitoring component. This component combines predefined ethical guidelines, based on established moral frameworks [23]–[25], with the AI's growing capacity for moral reasoning and its enhanced idea categorization capabilities.

The proposed framework has the potential to revolutionize AI development by creating self-improving systems that can generate, evaluate, and integrate new ideas while maintaining ethical alignment [26], [27]. However, implementing such a system also poses significant challenges, including ensuring the quality and relevance of generated ideas, balancing exploration and exploitation, measuring genuine self-improvement, and addressing scalability and computational requirements [28].

To address these challenges, we propose a set of strategies

and safeguards, such as rigorous evaluation metrics [29], human oversight [30], and collaboration between the AI system and human stakeholders [31]. By carefully designing and monitoring the self-improvement loop, we aim to create AI systems that can continuously improve while remaining transparent, accountable, and beneficial to society [32], [33].

The Gödel Machine, proposed by Jürgen Schmidhuber, is a theoretical framework for a self-improving AI system that uses a theorem prover to search for provably beneficial modifications to its own code [34]. In contrast, our framework takes a more dynamic and continuous approach to self-improvement by leveraging the generative and adaptive capabilities of large language models [1], [35]. This allows our system to generate novel ideas that are not only relevant and useful but also aligned with human cognitive processes. By focusing on optimizing for human understanding and creativity, our approach sets itself apart from the Gödel Machine and enables our framework to enhance human-AI collaboration and thrive in diverse and rapidly changing environments.

The rest of the paper is organized as follows: Section II provides an overview of related work and background concepts. Section III describes our proposed methodology in detail, including the idea generation, categorization, and integration processes, as well as the ethical alignment and monitoring component. Section IV discusses the potential benefits, challenges, and limitations of our approach, along with future research directions. Finally, Section V concludes the paper, emphasizing the importance of developing self-improving AI systems that align with human values and societal expectations.

II. Background

A. Continuous Learning in AI

Continuous learning is a crucial aspect of developing AI systems that can adapt to changing environments and acquire new knowledge over time. Various approaches have been proposed to enable continuous learning in AI, including online learning [36], lifelong learning [37], and meta-learning [38]. These approaches aim to overcome the limitations of traditional batch learning and enable AI systems to learn incrementally from new data without forgetting previously acquired knowledge.

B. Learning Paradigms

Several learning paradigms have been developed to optimize the learning process in AI systems. Curriculum Learning [12] presents training examples in a specific order, starting with easier examples and gradually progressing to more difficult ones. Self-Paced Learning [13] allows the model to control the order and speed at which it learns, focusing on the most informative or challenging examples. Active Learning [14] involves the model actively selecting the most informative examples to learn from, rather than passively accepting the provided examples.

C. Idea Generation and Creativity in AI

Idea generation and creativity are essential components of intelligent systems that can innovate and solve novel problems. Recent advancements in generative models, such as Generative Adversarial Networks (GANs) [8] and Variational Autoencoders (VAEs) [39], have enabled AI systems to generate new ideas and concepts in various domains, including art, music, and design [40]. However, evaluating the quality and usefulness of generated ideas remains a significant challenge [41].

D. Information Categorization and Retrieval

Efficient categorization and retrieval of information are crucial for AI systems to make informed decisions and generate relevant ideas. Techniques such as semantic networks [42], ontologies [43], and knowledge graphs [10] have been used to organize and represent knowledge in a structured manner. These approaches enable AI systems to understand the relationships between concepts and retrieve relevant information based on context [44].

E. Large Language Models (LLMs)

Large language models (LLMs) have demonstrated remarkable capabilities in various natural language processing tasks, including text generation, classification, and understanding [3], [4]. LLMs are trained on vast amounts of text data, enabling them to capture rich linguistic knowledge and generate coherent and contextually relevant text. While LLMs show promise for idea generation and categorization, their ability to produce truly novel and creative ideas or accurately categorize ideas based on specific criteria remains an open question [45], [46].

F. Few-Shot and Zero-Shot Learning

Few-shot and zero-shot learning techniques have emerged as powerful approaches for enabling AI systems to learn from limited examples or adapt to new tasks without explicit training data. Few-shot learning allows AI models to learn from a small number of labeled examples [47], while zero-shot learning enables models to perform tasks or recognize concepts they have not been directly trained on [48]. These techniques are particularly relevant for idea categorization, as they can help AI systems adapt to new idea categories or domains with minimal training data.

G. Domain-Specific Knowledge Integration

Incorporating domain-specific knowledge into AI systems is crucial for accurate and contextually relevant idea categorization and integration. Knowledge graphs [10] and ontologies [11] are powerful tools for representing and reasoning about domain-specific concepts and relationships. By integrating such structured knowledge into AI systems, we can enable more informed decision-making and improve the quality of generated and categorized ideas [49].

H. Self-Supervised Learning

Self-supervised learning is a paradigm in which AI systems learn to represent and understand data without explicit human annotations [50]. By leveraging the inherent structure and patterns in the data, self-supervised learning enables AI systems to acquire meaningful representations and improve their performance on downstream tasks [4]. This approach has been particularly successful in natural language processing and computer vision domains [51].

I. Incremental Learning and Knowledge Consolidation

Incremental learning and knowledge consolidation are essential for AI systems to continuously expand their knowledge base without forgetting previously acquired information. Techniques such as Experience Replay [52] and Gradient Episodic Memory [53] enable AI models to update their knowledge incrementally, while memory augmentation [21] and consistency regularization [22] help maintain the stability and coherence of the learned representations.

J. Ethical AI and Value Alignment

As AI systems become more autonomous and influential, ensuring their alignment with human values and ethical principles becomes paramount [31]. Various approaches have been proposed to incorporate ethics into AI systems, including value alignment [54], inverse reward design [55], and constrained optimization [56]. These approaches aim to create AI systems that make decisions consistent with human moral values and societal norms [28].

K. AI Governance and Oversight

Proper governance mechanisms and oversight processes are necessary to ensure the transparency, accountability, and ethical alignment of AI systems throughout their development and deployment. This includes establishing ethical guidelines, conducting impact assessments, and implementing regular audits and monitoring procedures [57]. Collaborative efforts among stakeholders, including researchers, policymakers, and industry practitioners, are crucial for developing effective governance frameworks that keep pace with the rapid advancements in AI technologies [58].

III. Methodology

A. Overview

Our proposed framework for self-improving AI systems revolves around a central LLM that continuously refines itself through a virtuous cycle of idea generation, categorization, and integration. Throughout this section, we will refer to the LLM as "the AI." The AI serves as the primary driver of the system, leveraging its inherent language understanding and generation capabilities to engage in a self-improvement process [3], [4].

The framework consists of three main components: idea generation, idea categorization, and idea integration. These components work together to enable the AI to generate novel ideas, classify and categorize them based on relevance and usefulness, and integrate the most promising ideas into its knowledge base. As the AI progresses through this self-improvement loop, it continuously enhances its abilities in each of these areas, leading to a powerful feedback effect [59], [60].

1) The Self-Refining AI: At the core of our framework is a single AI that undergoes a continuous process of self-refinement. As the AI generates new ideas, categorizes them, and integrates the most valuable ones into its knowledge base, it gradually improves its performance in each of these tasks. This self-reinforcing process allows the AI to bootstrap its own intelligence, becoming more adept at generating high-quality ideas, accurately categorizing them, and seamlessly integrating them into its existing knowledge [1], [35].

The AI's self-improvement is driven by the feedback it receives from the idea categorization component, which assesses the quality and relevance of the generated ideas. By learning from this feedback and adjusting its internal representations and parameters accordingly, the AI becomes increasingly proficient at generating ideas that are both novel and valuable, while also improving its ability to categorize and integrate them effectively [9], [61].

2) Interaction of Components: The three components of the framework - idea generation, idea categorization, and idea integration - work in synergy to enable the AI's self-improvement. The idea generation component produces a diverse range of ideas by leveraging the AI's language generation capabilities and incorporating techniques such as prompt engineering and sampling methods [5], [6]. These generated ideas are then processed by the idea categorization component, which adapts the AI's language understanding capabilities to assess the relevance, usefulness, and other predefined criteria of the ideas [9].

The most promising ideas, as determined by the categorization component, are then passed to the idea integration component. This component focuses on incorporating the selected ideas into the AI's knowledge base while maintaining consistency and coherence [20], [21]. As the AI integrates new knowledge, it enhances its ability to generate more sophisticated and relevant ideas in subsequent iterations, creating a positive feedback loop that drives its continuous self-improvement [59], [60].

Throughout this process, the AI's performance in idea generation, categorization, and integration steadily improves, enabling it to tackle increasingly complex and open-ended problems. By leveraging the power of self-refinement and the synergistic interaction of the three components, our framework aims to create an AI system that can autonomously learn, adapt, and innovate, pushing the boundaries of what is possible with artificial intelligence [1], [35].

B. Idea Generation

Building upon the AI's language generation capabilities, the idea generation component employs various techniques and tools to produce novel and diverse ideas. These techniques and tools serve as "training wheels" or scaffolding that support the

AI's idea generation process, particularly in the early stages of its self-improvement journey.

1) Prompt Engineering: One of the key techniques used to guide the AI's idea generation process is prompt engineering [5]. By carefully designing prompts that provide context, constraints, or inspiration, we can steer the AI towards exploring specific domains or concepts. As the AI progresses through the self-improvement loop, it becomes increasingly adept at understanding and responding to these prompts, generating ideas that are more relevant and coherent.

However, as the AI's knowledge base expands and its language understanding capabilities improve, it becomes less reliant on explicit prompts. Instead, it can generate high-quality ideas more autonomously, drawing upon its internalized knowledge and understanding of the world [1], [35].

2) Sampling Techniques: To encourage the exploration of diverse and creative ideas, we employ various sampling techniques during the idea generation process. Beam search, top-k sampling, and nucleus sampling are among the techniques used to guide the AI's generation process [6].

These sampling techniques allow the AI to consider multiple possible paths or outcomes when generating text, promoting the creation of unique and unexpected ideas. As the AI refines itself through the self-improvement loop, it learns to internalize these sampling strategies, becoming more effective at generating diverse and creative ideas without explicit guidance.

3) Specialized Generative Models: Specialized generative models, such as VAEs [7] and GANs [8], are employed as tools to enhance the AI's idea generation capabilities, particularly in the realm of non-textual content. By conditioning these models on the AI's outputs, we can create a symbiotic relationship between the AI and these tools [40].

Initially, the AI relies heavily on these specialized models to generate non-textual content that complements its text-based ideas. However, as the AI's understanding of the relationships between concepts and their non-textual representations improves, it becomes more adept at generating high-level concepts or descriptions that effectively guide these specialized models. This progression reduces the AI's dependence on these tools over time, as it learns to generate more comprehensive and coherent ideas autonomously.

4) Iterative Refinement: The idea generation component is an iterative process that continuously refines itself based on feedback from the idea categorization component. As the AI generates ideas and receives feedback on their quality and relevance, it adjusts its parameters, prompts, and training data accordingly [59], [60].

This feedback loop allows the AI to learn from its successes and failures, prioritizing the generation of ideas that align with the criteria established by the categorization component. Over time, the AI internalizes this feedback, becoming more effective at generating high-quality ideas without explicit guidance from the categorization component.

C. Idea Categorization

The idea categorization component plays a crucial role in the self-improvement loop by assessing the quality, relevance, and potential impact of the ideas generated by the AI. This component draws inspiration from human cognition, specifically our ability to quickly categorize information based on its perceived value, such as distinguishing between useful and irrelevant ideas or identifying the most promising concepts for further exploration [62].

1) Adapting LLMs for Idea Categorization: To enable the AI to perform idea categorization effectively, we employ various techniques that leverage its language understanding capabilities. Zero-shot learning, few-shot learning, and supervised fine-tuning are among the techniques used to adapt the AI to the specific task of categorizing ideas based on their semantic content and relevance to the target domain [9].

Initially, the AI may rely heavily on these techniques to acquire the necessary knowledge and skills for accurate idea categorization. However, as the AI progresses through the self-improvement loop and integrates new knowledge, it becomes increasingly adept at categorizing ideas without explicit guidance or training. This progression mimics the way humans become more efficient at categorizing information as they acquire expertise in a particular domain [63].

2) Incorporating Domain-Specific Knowledge: To further enhance the AI's idea categorization capabilities, we incorporate domain-specific knowledge through the use of knowledge graphs [10] and ontologies [11]. These tools provide structured representations of concepts, relationships, and hierarchies within a given domain, enabling the AI to develop a more nuanced understanding of the context in which ideas are generated and evaluated.

By integrating domain-specific knowledge into the categorization process, the AI can make more informed and context-aware judgments about the relevance and potential impact of ideas. This integration also allows the AI to draw connections between seemingly disparate concepts, a key aspect of human creativity and problem-solving [64].

3) Assessing Idea Quality and Potential Impact: The idea categorization component assesses the quality and potential impact of generated ideas across multiple dimensions, such as novelty, feasibility, scalability, and alignment with user preferences and needs. By considering these factors, the AI can prioritize ideas that have the greatest potential to drive meaningful advancements and create value for users.

As the AI refines its idea categorization capabilities through the self-improvement loop, it becomes more adept at recognizing patterns, identifying promising ideas, and making swift, intuitive judgments about their potential impact. This progression mirrors the way human experts develop a keen sense of what constitutes a valuable or impactful idea within their domain of expertise [65].

4) Handling Ideas of Different Sizes: Ideas can vary in size and granularity, ranging from a single word to a sentence, a book, or a summary of a book. Determining the ideal "idea size" can be challenging, as it depends on the specific context,

domain, and objectives of the self-improving AI system. To address this challenge, we propose a nuanced approach that includes:

1) Hierarchical idea representation: Implementing a hierarchical structure for representing ideas can help manage the varying granularity of ideas, ranging from high-level concepts to more specific and detailed ideas [66].
2) Adaptive idea segmentation: Developing techniques for adaptively segmenting text into meaningful idea units based on the context and domain can help the system handle ideas of different sizes effectively [67].
3) Multi-scale idea processing: Designing the idea generation, categorization, and integration components to operate at multiple scales can enable the system to process ideas of different sizes effectively, using techniques such as hierarchical attention mechanisms and multi-resolution representations [68].
4) Domain-specific idea size optimization: Incorporating domain knowledge and conducting empirical studies to determine the optimal idea size for specific applications can help tailor the system's handling of idea sizes to the particular requirements of each domain [69].

By incorporating these strategies, the idea categorization component can become more nuanced and effective in handling ideas of different sizes, enabling the self-improving AI system to generate, evaluate, and integrate ideas at the most appropriate level of granularity for each specific context.

5) Continuous Refinement and Adaptation: The idea categorization component is not a static entity but rather a dynamic and evolving process that continuously refines itself based on the AI's interactions with the idea generation and integration components. As the AI generates new ideas and integrates the most promising ones into its knowledge base, it gains a more comprehensive understanding of the factors that contribute to idea quality and impact.

This continuous refinement and adaptation process allows the AI to develop increasingly sophisticated and nuanced categorization capabilities over time. By learning from its successes and failures, the AI can adjust its categorization criteria and decision-making processes, becoming more effective at identifying and prioritizing high-potential ideas.

6) Importance-Weighted Learning: Importance-Weighted learning is a novel learning paradigm that builds upon the idea categorization component's ability to assess the quality, relevance, and potential impact of generated ideas [41], [64]. It leverages the categorization process to prioritize the most valuable and informative examples in the training data [12]–[14], optimizing the learning process by focusing on the most critical aspects. By assigning higher importance to examples deemed more relevant, impactful, and aligned with the system's objectives, importance-weighted learning enables the AI system to learn more efficiently and effectively [15]–[17].

The idea categorization component can be extended to preprocess and label any type of training data [9], [20]. Applying categorization techniques to input data allows the system to assign value-based labels to each example, indicating its importance and relevance to the learning process. This ensures consistency between idea generation and learning processes, enables focus on the most valuable examples, and facilitates the integration of domain-specific knowledge and ethical considerations into data labeling [10], [56].

Importance-weighted learning is applicable to various data types, such as text, images, or structured data. For textual data, the idea categorization component assesses semantic relevance, coherence, and potential impact, assigning higher value labels to the most informative and meaningful examples [4], [61]. Similarly, for image data, the categorization component evaluates visual quality, diversity, and representativeness, prioritizing the most valuable examples for training [40], [51].

Implementing importance-weighted learning within the Idea Categorization framework involves techniques such as importance sampling [15], dynamic loss weighting [16], and adaptive data selection [17]. These techniques enable the system to prioritize the most valuable examples during the learning process based on the value labels assigned by the categorization component. Integrating these techniques into the idea categorization pipeline allows the AI system to continuously refine its learning process, concentrating on the most impactful and relevant aspects of the training data [12], [13].

The integration of importance-weighted learning into the Idea Categorization process offers several benefits. It enables more efficient learning by focusing on the most informative and relevant examples, reducing the computational resources required for training. Moreover, it promotes alignment between the learning process and the system's objectives, ensuring that the acquired knowledge is directly relevant to the desired outcomes. Finally, it facilitates the incorporation of domain-specific knowledge and ethical considerations into the learning process, resulting in more accurate, context-aware, and value-aligned AI systems.

D. Idea Integration

The idea integration component is responsible for incorporating the most promising ideas, as identified by the idea categorization component, into the AI's knowledge base. This process is crucial for the AI's continuous self-improvement, as it allows the model to expand its understanding of the world, refine its reasoning capabilities, and generate increasingly sophisticated and impactful ideas.

1) Self-Querying as a Fundamental Tool: At the core of the idea integration process is the AI's ability to engage in self-querying, a natural and intuitive tool for assessing the value and compatibility of new ideas with its existing knowledge base. By comparing newly generated ideas or concepts with its current understanding, the AI can determine whether integrating these ideas would lead to a meaningful improvement in its knowledge and capabilities [18], [19].

Self-querying allows the AI to ask itself questions such as: "Does this new idea provide a more comprehensive or nuanced understanding of a particular concept?", "How does this idea

relate to my existing knowledge, and does it fill any gaps or resolve any inconsistencies?", or "Can this idea be applied to solve problems or generate insights in ways that my current knowledge cannot?". By engaging in this internal dialogue, the AI can make informed decisions about which ideas to integrate and how to incorporate them into its knowledge base effectively.

2) Fine-Tuning and Adaptation: To facilitate the seamless integration of new ideas, the AI undergoes continuous fine-tuning and adaptation. Techniques such as incremental fine-tuning [20], memory augmentation [21], and consistency regularization [22] are employed to gradually update the AI's parameters and representations, ensuring that new knowledge is incorporated without overwriting or disrupting existing knowledge.

As the AI integrates new ideas, it must also adapt its language understanding and generation capabilities to effectively utilize this expanded knowledge base. Fine-tuning allows the AI to refine its ability to interpret and generate text that incorporates newly acquired concepts and ideas, enabling it to engage in more sophisticated and context-aware communication.

3) Knowledge Base Maintenance and Consistency: Maintaining the consistency and coherence of the AI's knowledge base is a critical aspect of the idea integration process. As new ideas are incorporated, the AI must ensure that they are compatible with its existing knowledge and do not introduce logical inconsistencies or contradictions.

To address this challenge, the AI employs various techniques, such as knowledge graph embedding [70], ontology alignment [71], and consistency checking [72], to monitor and maintain the integrity of its knowledge base. These tools allow the AI to identify and resolve any inconsistencies that may arise during the idea integration process, ensuring that its knowledge remains logically sound and coherent.

4) Incremental Learning and Consolidation: Idea integration is an incremental and iterative process, where the AI continuously builds upon its existing knowledge and refines its understanding over time. As new ideas are integrated, the AI engages in a process of consolidation, where it strengthens the connections between related concepts, identifies patterns and relationships, and develops a more comprehensive and unified understanding of the world.

Incremental learning algorithms, such as Experience Replay [52] and Gradient Episodic Memory [53], play a crucial role in this consolidation process, allowing the AI to efficiently update its knowledge base while preserving previously acquired knowledge. By continuously revisiting and reinforcing integrated ideas, the AI can develop a more robust and interconnected knowledge base that supports its ongoing self-improvement.

5) Retraining for Holistic Idea Integration: Retraining the AI on an updated dataset that incorporates newly generated ideas is another approach to idea integration. This method involves updating the AI's training data with the most promising ideas, along with their associated labels or metadata, and retraining the model from the beginning. This allows the model to learn to integrate the new ideas into its knowledge base in a more holistic and unified manner [37], [73]. Retraining is particularly effective when the newly generated ideas represent a significant shift or expansion in the AI's knowledge domain. In such cases, applying importance-weighted learning techniques during retraining enables the AI to develop a more coherent understanding of the new ideas in the context of its existing knowledge, ensuring a more seamless integration process while maintaining consistency and alignment with the AI's intended purpose and values [74], [75].

E. Ethical Alignment and Monitoring

The ethical alignment and monitoring component is a critical element of the AI's self-improvement loop, ensuring that the model's growth and development remain firmly grounded in human values and ethical principles. This component combines predefined ethical guidelines, based on established moral frameworks [23]–[25], with the AI's growing capacity for moral reasoning and its enhanced idea categorization capabilities.

1) Promoting Ethical Idea Integration: During the idea evaluation and integration process, the AI employs a combination of predefined ethical guidelines, encoded using formal logic and constraint satisfaction techniques [76], and dynamic ethical reasoning to identify and exclude ideas that violate human values and moral principles, while actively promoting ideas that are ethically sound. As the AI's intelligence and knowledge base expand through the self-improvement loop, it develops an increasing capacity for moral reasoning, considering the context, intent, and potential consequences of ideas using techniques such as counterfactual reasoning [77], analogical reasoning [78], and multi-objective optimization [56].

The AI's ethical reasoning capabilities also inform the prioritization and selection of ideas during the categorization phase, ensuring that the ideas ultimately integrated into its knowledge base are not only innovative and impactful but also ethically sound. Continuous learning and updating of the ethical guidelines and reasoning capabilities, through ongoing feedback from human experts, regular audits, and active learning from real-world interactions, ensure the robustness and adaptability of the ethical alignment component.

2) Generating Context-Aware and Value-Aligned Outputs: The AI leverages its growing proficiency in idea categorization to generate context-aware and value-aligned outputs. By accurately categorizing ideas based on their semantic content, intelligence, usefulness, and ethical implications, the AI can prioritize and filter its outputs to ensure they are informative, relevant, and aligned with human values and social norms. This involves employing techniques from algorithmic fairness [79], explainable AI [80], and counterfactual fairness metrics [81] to detect and mitigate potential biases and fairness issues, while generating human-interpretable explanations of its reasoning [82].

3) Integration with the Self-Improvement Loop and Human Collaboration: The ethical alignment and monitoring component is deeply integrated into the AI's self-improvement loop, influencing and guiding the model's growth and development at every stage. As the AI's knowledge base expands and its intelligence increases, the ethical alignment component continuously adapts and grows alongside it, enabling more sophisticated moral reasoning and a nuanced understanding of the implications of its actions.

Close collaboration between the AI and human stakeholders, including researchers, ethicists, policymakers, and the public, is essential for ensuring that the AI's ethical framework remains aligned with human values and societal norms. Regular dialogue and consultation allow the AI to gain insights into the ethical concerns and priorities of the communities it serves, informing the ongoing refinement of its ethical guidelines and reasoning capabilities, and fostering public trust and confidence in the AI's ethical alignment.

IV. Discussion

A. Potential Benefits

The proposed framework for self-improving AI systems has the potential to bring about significant advancements in AI capabilities and applications while addressing the critical issue of AI alignment. By enabling AI to generate, evaluate, and integrate new ideas continuously, this approach can lead to more creative, adaptable, and efficient AI systems that are better aligned with human values and societal expectations [1].

1) Addressing the AI Alignment Problem: While AIs can generate coherent text, their focus on next-token prediction does not ensure alignment with human values or ethics [61]. Despite potentially understanding what is considered good or bad based on their training data, AIs lack explicit value alignment, leading to the potential generation of harmful or misaligned text [28], [31]. This misalignment poses significant risks as AI systems become more advanced and influential in various domains.

The proposed framework addresses this critical issue by introducing a structured approach to idea categorization that actively incorporates value alignment into the training process. By explicitly training AI systems to classify ideas based on their alignment with ethical principles and societal norms, the proposed approach aims to create AI systems that are more aware of and aligned with human values, while still leveraging the generative capabilities of AIs and specialized models [35]. This value-aligned idea categorization approach represents a significant step towards developing AI systems that are not only capable of generating coherent text but also more ethically aligned with human values, mitigating the potential dangers posed by current AI systems.

2) Enhanced Creativity and Innovation: The ability to generate and explore a vast space of ideas can enable AI systems to discover novel solutions and approaches that may not be apparent to human experts [41]. This enhanced creativity can lead to breakthrough innovations in various domains, from scientific research to artistic expression [40]. By leveraging the power of idea generation and categorization, the proposed framework can unlock the creative potential of AI systems while ensuring that this creativity is directed towards beneficial outcomes aligned with human values.

3) Impact on Various Industries and Domains: Self-improving AI systems have the potential to revolutionize various industries and domains, such as healthcare, education, finance, and transportation. In healthcare, these systems could assist in drug discovery, personalized medicine, and clinical decision support [83]. In education, self-improving AI could enable adaptive learning, intelligent tutoring systems, and automated content creation [84]. In finance, these systems could enhance risk assessment, fraud detection, and algorithmic trading [85]. By focusing on ethical alignment and the generation of beneficial ideas, the proposed framework could serve as a foundation for developing domain-specific self-improving AI applications that drive innovation and efficiency across various sectors while ensuring that these advancements are aligned with human values and societal expectations.

B. Challenges and Limitations

The proposed framework for self-improving AI systems presents several challenges and limitations that are unique and relevant to the specific methods discussed in this paper. These challenges must be addressed to ensure the framework's effective and responsible implementation.

1) Balancing Exploration and Exploitation in Idea Generation: Striking the right balance between exploration and exploitation is crucial for the idea generation component of the framework, which aims to produce novel and diverse ideas. Overemphasis on either exploration or exploitation can lead to suboptimal outcomes, such as the generation of impractical ideas or getting stuck in suboptimal solution spaces [86]. Developing adaptive strategies that dynamically adjust this trade-off based on the system's performance and problem complexity is an ongoing research challenge specific to the idea generation component of the self-improving AI framework [87].

2) Robustness and Generalization in Idea Categorization: Ensuring the robustness and generalization capabilities of the idea categorization component, which relies on AIs and domain-specific knowledge integration, is essential for accurately categorizing novel ideas in real-world scenarios. The system must handle variations in idea representation, noise in the input data, and potential adversarial attacks that could manipulate the categorization process [88]. For example, an adversary could generate ideas with subtle semantic variations that exploit vulnerabilities in the AIs, leading to misclassification. Few-shot learning, data augmentation, and adversarial training can improve the robustness of the categorization models, but further research is needed to guarantee reliable performance in the context of the self-improving AI framework.

3) Maintaining Consistency and Coherence in Idea Integration: As the AI system's knowledge base grows through the

integration of new ideas, ensuring its logical consistency and semantic coherence becomes increasingly challenging [10]. Techniques such as ontology alignment, knowledge graph embedding, and consistency checking are particularly relevant to the idea integration component of the framework, as they help maintain the integrity and reliability of the knowledge base over time. However, the unique challenges posed by the continuous integration of novel ideas generated by the self-improving AI system require the development of scalable and efficient methods tailored to this specific context.

4) Ethical Alignment and Monitoring: Ensuring that the self-improving AI system remains aligned with human values and ethical principles throughout its learning and adaptation process is a critical challenge addressed by the ethical alignment and monitoring component of the framework [31]. Developing a comprehensive and universally acceptable set of ethical principles for AI systems is a complex and ongoing challenge that requires input from multiple stakeholders [89]. The inherent subjectivity and context-dependence of ethical considerations make it difficult to formalize and embed these principles into the AI system's decision-making processes [90]. Designing effective monitoring and intervention mechanisms that can detect and prevent unintended or unethical behavior in real-time is an open research problem that is particularly relevant to the self-improving AI framework, given its autonomous learning and adaptation capabilities [91].

5) Transitioning to Secure and Decentralized Infrastructure: Ensuring the security and tamper-proofness of the self-improving AI system's critical components, such as its alignment mechanisms and governance processes, is a significant challenge that requires the adoption of novel architectural solutions. Decentralized AI architectures based on blockchain technology and on-chain AI smart contracts have emerged as a promising approach to address this challenge by leveraging the inherent immutability and transparency of distributed ledger systems [92]. However, transitioning to these secure and decentralized infrastructures presents its own set of obstacles, as it necessitates a fundamental shift in the design, development, and governance paradigms of AI systems [93].

C. Future Research Directions

The self-improving AI framework presented in this paper opens up several avenues for future research:

1) Implementing and testing the proposed framework in practical settings to validate its feasibility, scalability, and performance.
2) Conducting empirical studies to evaluate the effectiveness of the proposed techniques for balancing exploration and exploitation, ensuring robustness and generalization, maintaining consistency and coherence, and aligning with ethical principles.
3) Exploring the integration of the self-improving AI framework with existing AI architectures and platforms to assess its compatibility and potential for synergistic improvements.
4) Engaging with the broader AI community, domain experts, policymakers, and the public to gather feedback and collaborate on the development of guidelines and evaluation criteria for self-improving AI systems.

These research directions will contribute to the practical realization and refinement of the self-improving AI framework, paving the way for more capable, robust, and ethically aligned AI systems.

REFERENCES

[1] B. M. Lake, T. D. Ullman, J. B. Tenenbaum, and S. J. Gershman, "Building machines that learn and think like people," *Behavioral and brain sciences*, vol. 40, 2017.

[2] G. Marcus, "Deep learning: A critical appraisal," *arXiv preprint arXiv:1801.00631*, 2018.

[3] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell *et al.*, "Language models are few-shot learners," *arXiv preprint arXiv:2005.14165*, 2020.

[4] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," *arXiv preprint arXiv:1810.04805*, 2019.

[5] P. Liu, W. Yuan, J. Fu, Z. Jiang, H. Hayashi, and G. Neubig, "Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing," *arXiv preprint arXiv:2107.13586*, 2021.

[6] A. Holtzman, J. Buys, L. Du, M. Forbes, and Y. Choi, "The curious case of neural text degeneration," *arXiv preprint arXiv:1904.09751*, 2020.

[7] D. P. Kingma and M. Welling, "Auto-encoding variational bayes," *arXiv preprint arXiv:1312.6114*, 2013.

[8] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," in *Advances in neural information processing systems*, 2014, pp. 2672–2680.

[9] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu, "Exploring the limits of transfer learning with a unified text-to-text transformer," in *Journal of Machine Learning Research*, vol. 21, no. 140, 2020, pp. 1–67.

[10] S. Ji, S. Pan, E. Cambria, P. Marttinen, and S. Y. Philip, "A survey on knowledge graphs: Representation, acquisition, and applications," *IEEE Transactions on Neural Networks and Learning Systems*, 2021.

[11] N. Guarino, "Formal ontology and information systems," in *Formal Ontology in Information Systems: Proceedings of the First International Conference (FOIS'98), June 6-8, Trento, Italy*. IOS Press, 1998, pp. 3–15.

[12] Y. Bengio, J. Louradour, R. Collobert, and J. Weston, "Curriculum learning," in *International Conference on Machine Learning*, 2009, pp. 41–48.

[13] M. P. Kumar, B. Packer, and D. Koller, "Self-paced learning for latent variable models," in *Advances in Neural Information Processing Systems*, 2010, pp. 1189–1197.

[14] B. Settles, "Active learning literature survey," *Computer Sciences Technical Report 1648, University of Wisconsin–Madison*, 2009.

[15] A. Katharopoulos and F. Fleuret, "Not all samples are created equal: Deep learning with importance sampling," in *International Conference on Machine Learning*, 2018, pp. 2525–2534.

[16] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Doll'ar, "Focal loss for dense object detection," in *IEEE International Conference on Computer Vision*, 2017, pp. 2980–2988.

[17] Y. Fan, F. Tian, T. Qin, X.-Y. Li, and T.-Y. Liu, "Learning to active learn: A deep reinforcement learning approach," in *Conference on Empirical Methods in Natural Language Processing*, 2017, pp. 595–605.

[18] F. Petroni, T. Rockt"aschel, S. Riedel, P. Lewis, A. Bakhtin, Y. Wu, and A. Miller, "Language models as knowledge bases?" in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 2463–2473.

[19] Z. Jiang, F. F. Xu, J. Araki, and G. Neubig, "How can we know what language models know?" in *Transactions of the Association for Computational Linguistics*, vol. 8. MIT Press, 2020, pp. 423–438.

[20] S. Gururangan, A. Marasovi'c, S. Swayamdipta, K. Lo, I. Beltagy, D. Downey, and N. A. Smith, "Don't stop pretraining: Adapt language models to domains and tasks," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020, pp. 8342–8360.

[21] J. W. Rae, A. Potapenko, S. M. Jayakumar, C. Hillier, and T. P. Lillicrap, "Compressive transformers for long-range sequence modelling," *arXiv preprint arXiv:1911.05507*, 2019.

[22] W. Liang, J. Zou, and Z. Yu, "Alice: Active learning with contrastive natural language explanations," *arXiv preprint arXiv:2009.10259*, 2020.

[23] J. S. Mill, *Utilitarianism*. Parker, Son and Bourn, 1863.

[24] I. Kant, *Groundwork of the Metaphysics of Morals*. Oxford University Press, 1785.

[25] Aristotle, *Nicomachean ethics*. Hackett Publishing, 1999.

[26] R. Bommasani, D. A. Hudson, E. Adeli, R. Altman, S. Arora, S. von Arx, M. S. Bernstein, J. Bohg, A. Bosselut, E. Brunskill *et al.*, "On the opportunities and risks of foundation models," *arXiv preprint arXiv:2108.07258*, 2021.

[27] D. Silver, S. Singh, D. Precup, and R. S. Sutton, "Reward is enough," *Artificial Intelligence*, vol. 299, p. 103535, 2021.

[28] N. Bostrom, *Superintelligence: Paths, dangers, strategies*. Oxford University Press, 2017.

[29] D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt, "Measuring massive multitask language understanding," *arXiv preprint arXiv:2009.03300*, 2020.

[30] S. Amershi, D. Weld, M. Vorvoreanu, A. Fourney, B. Nushi, P. Collisson, J. Suh, S. Iqbal, P. N. Bennett, K. Inkpen *et al.*, "Guidelines for human-ai interaction," *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, pp. 1–13, 2019.

[31] S. Russell, *Human compatible: Artificial intelligence and the problem of control*. Penguin, 2019.

[32] L. Floridi and J. Cowls, "Establishing the rules for building trustworthy ai," *Nature Machine Intelligence*, vol. 1, no. 6, pp. 261–262, 2019.

[33] A. Jobin, M. Ienca, and E. Vayena, "The global landscape of ai ethics guidelines," *Nature Machine Intelligence*, vol. 1, no. 9, pp. 389–399, 2019.

[34] J. Schmidhuber, "Gödel machines: Fully self-referential optimal universal self-improvers," in *Artificial general intelligence*. Springer, 2007, pp. 199–226.

[35] Y. Bengio, "Consciousness prior," *arXiv preprint arXiv:1709.08568*, 2017.

[36] S. C. H. Hoi, D. Sahoo, J. Lu, and P. Zhao, "Online learning: A comprehensive survey," *arXiv preprint arXiv:1802.02871*, 2018.

[37] G. I. Parisi, R. Kemker, J. L. Part, C. Kanan, and S. Wermter, "Continual lifelong learning with neural networks: A review," *Neural Networks*, vol. 113, pp. 54–71, 2019.

[38] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *International Conference on Machine Learning*. PMLR, 2017, pp. 1126–1135.

[39] D. P. Kingma and M. Welling, "Auto-encoding variational bayes," *arXiv preprint arXiv:1312.6114*, 2014.

[40] A. Ramesh, M. Pavlov, G. Goh, S. Gray, C. Voss, A. Radford, M. Chen, and I. Sutskever, "Zero-shot text-to-image generation," *arXiv preprint arXiv:2102.12092*, 2021.

[41] M. A. Boden, *The creative mind: Myths and mechanisms*. Routledge, 2004.

[42] J. F. Sowa, "Semantic networks," *Encyclopedia of Cognitive Science*, vol. 4, pp. 1–6, 2006.

[43] T. R. Gruber, "A translation approach to portable ontology specifications," *Knowledge acquisition*, vol. 5, no. 2, pp. 199–220, 1993.

[44] A. Bordes, N. Usunier, A. Garcia-Duran, J. Weston, and O. Yakhnenko, "Translating embeddings for modeling multi-relational data," in *Advances in neural information processing systems*, 2013, pp. 2787–2795.

[45] E. M. Bender, T. Gebru, A. McMillan-Major, and S. Shmitchell, "On the dangers of stochastic parrots: Can language models be too big," *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, pp. 610–623, 2021.

[46] G. Marcus, "The next decade in ai: four steps towards robust artificial intelligence," *arXiv preprint arXiv:2002.06177*, 2020.

[47] Y. Wang, Q. Yao, J. T. Kwok, and L. M. Ni, "Generalizing from a few examples: A survey on few-shot learning," in *ACM Computing Surveys (CSUR)*, vol. 53, no. 3. ACM New York, NY, USA, 2020, pp. 1–34.

[48] Y. Xian, C. H. Lampert, B. Schiele, and Z. Akata, "Zero-shot learninga comprehensive evaluation of the good, the bad and the ugly," *IEEE transactions on pattern analysis and machine intelligence*, vol. 41, no. 9, pp. 2251–2265, 2018.

[49] A. Bosselut, H. Rashkin, M. Sap, C. Malaviya, A. Celikyilmaz, and Y. Choi, "Comet: Commonsense transformers for automatic knowledge graph construction," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 2019, pp. 4762–4779.

[50] L. Jing and Y. Tian, "Self-supervised visual feature learning with deep neural networks: A survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2020.

[51] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *International conference on machine learning*. PMLR, 2020, pp. 1597–1607.

[52] D. Rolnick, A. Ahuja, J. Schwarz, T. Lillicrap, and G. Wayne, "Experience replay for continual learning," *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[53] D. Lopez-Paz and M. Ranzato, "Gradient episodic memory for continual learning," in *Advances in Neural Information Processing Systems*, 2017, pp. 6467–6476.

[54] N. Soares, "Value learning for artificial intelligence," *Machine Intelligence Research Institute*, 2016.

[55] D. Hadfield-Menell, S. Milli, P. Abbeel, S. J. Russell, and A. Dragan, "Inverse reward design," in *Advances in neural information processing systems*, 2017, pp. 6765–6774.

[56] R. Noothigattu, D. Bouneffouf, N. Mattei, R. Chandra, P. Madan, K. Varshney, M. Campbell, F. Rossi, E. Breckenridge, and S. Agarwal, "Interpretable multi-objective reinforcement learning through policy orchestration," *arXiv preprint arXiv:1809.08343*, 2018.

[57] C. Cath, S. Wachter, B. Mittelstadt, M. Taddeo, and L. Floridi, "Artificial intelligence and the 'good society': the us, eu, and uk approach," *Science and engineering ethics*, vol. 24, no. 2, pp. 505–528, 2018.

[58] L. Floridi, J. Cowls, M. Beltrametti, R. Chatila, P. Chazerand, V. Dignum, C. Luetge, R. Madelin, U. Pagallo, F. Rossi *et al.*, "Ai4peoplean ethical framework for a good ai society: opportunities, risks, principles, and recommendations," *Minds and Machines*, vol. 28, no. 4, pp. 689–707, 2018.

[59] A. Wang, Y. Pruksachatkun, N. Nangia, A. Singh, J. Michael, F. Hill, O. Levy, and S. Bowman, "Superglue: A stickier benchmark for general-purpose language understanding systems," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[60] D. Kiela, M. Bartolo, Y. Nie, D. Kaushik, A. Geiger, Z. Wu, B. Vidgen, G. Prasad, A. Singh, P. Ringshia *et al.*, "Dynabench: Rethinking benchmarking in nlp," in *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2021, pp. 4110–4124.

[61] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, "Language models are unsupervised multitask learners," 2019.

[62] D. Kahneman, "Thinking, fast and slow," 2011.

[63] W. G. Chase and H. A. Simon, "Perception in chess," *Cognitive Psychology*, vol. 4, no. 1, pp. 55–81, 1973.

[64] M. Benedek, B. Borovnjak, A. C. Neubauer, and S. Kruse-Weber, "Creativity and personality in classical, jazz and folk musicians," *Personality and Individual Differences*, vol. 52, no. 4, pp. 485–489, 2012.

[65] K. A. Ericsson, R. T. Krampe, and C. Tesch-R"omer, "The role of deliberate practice in the acquisition of expert performance," *Psychological Review*, vol. 100, no. 3, p. 363, 1993.

[66] Y. Shen, S. Tan, A. Sordoni, and A. Courville, "A hierarchical representation for language processing and reasoning," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 2019, pp. 6300–6309.

[67] A. A. Alemi, F. Chollet, G. Irving, C. Szegedy, and J. Urban, "Improving language understanding by generative pre-training," *arXiv preprint arXiv:1810.04805*, 2018.

[68] Z. Yang, D. Yang, C. Dyer, X. He, A. Smola, and E. Hovy, "Hierarchical attention networks for document classification," in *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2016, pp. 1480–1489.

[69] Y. Guo, Y. Ge, A. O. V. de Vries, and G. de Vries, "Domain-specific language model pretraining for biomedical natural language processing," *arXiv preprint arXiv:1909.03087*, 2019.

[70] Q. Wang, Z. Mao, B. Wang, and L. Guo, "Knowledge graph embedding: A survey of approaches and applications," *IEEE Transactions on Knowledge and Data Engineering*, vol. 29, no. 12, pp. 2724–2743, 2017.

[71] J. Euzenat and P. Shvaiko, *Ontology matching*. Springer, 2013.

[72] Z. Li and L. Specia, "Consistency of a recurrent language model with respect to incomplete decoding," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 5553–5562.

[73] J. L. McClelland, B. L. McNaughton, and R. C. O'Reilly, "Why there are complementary learning systems in the hippocampus and neocortex: Insights from the successes and failures of connectionist models of learning and memory," in *Psychological Review*, vol. 102, no. 3. American Psychological Association, 1995, pp. 419–457.

[74] Z. Chen and B. Liu, "Lifelong machine learning," *Synthesis Lectures on Artificial Intelligence and Machine Learning*, vol. 12, no. 3, pp. 1–207, 2018.

[75] F.-K. Sun, C.-H. Ho, and H.-Y. Lee, "Lamol: Language modeling for lifelong language learning," *arXiv preprint arXiv:2004.08071*, 2020.

[76] F. Rossi, P. Van Beek, and T. Walsh, *Handbook of constraint programming*. Elsevier, 2006.

[77] E. Perez, F. Strub, H. De Vries, V. Dumoulin, and A. Courville, "Learning to generate counterfactual explanations," in *Proceedings of the 33rd AAAI Conference on Artificial Intelligence*, 2019, pp. 4504–4511.

[78] M. Dehghani, E. Tomai, K. Forbus, and M. Klenk, "An integrated reasoning approach to moral decision-making," in *Proceedings of the 23rd AAAI Conference on Artificial Intelligence*, 2008, pp. 1280–1286.

[79] N. Mehrabi, F. Morstatter, N. Saxena, K. Lerman, and A. Galstyan, "A survey on bias and fairness in machine learning," *ACM Computing Surveys (CSUR)*, vol. 54, no. 6, pp. 1–35, 2021.

[80] A. Adadi and M. Berrada, "Peeking inside the black-box: A survey on explainable artificial intelligence (xai)," *IEEE Access*, vol. 6, pp. 52 138–52 160, 2018.

[81] M. J. Kusner, J. Loftus, C. Russell, and R. Silva, "Counterfactual fairness," in *Advances in Neural Information Processing Systems*, 2017, pp. 4066–4076.

[82] S. Wachter, B. Mittelstadt, and C. Russell, "Counterfactual explanations without opening the black box: Automated decisions and the gdpr," *Harvard Journal of Law & Technology*, vol. 31, no. 2, pp. 841–887, 2018.

[83] E. J. Topol, "High-performance medicine: the convergence of human and artificial intelligence," *Nature medicine*, vol. 25, no. 1, pp. 44–56, 2019.

[84] B. P. Woolf, *Building intelligent interactive tutors: Student-centered strategies for revolutionizing e-learning*. Morgan Kaufmann, 2010.

[85] A. Bahrammirzaee, "A comparative survey of artificial intelligence applications in finance: artificial neural networks, expert system and hybrid intelligent systems," *Neural Computing and Applications*, vol. 19, no. 8, pp. 1165–1195, 2010.

[86] R. S. Sutton and A. G. Barto, *Reinforcement learning: An introduction*. MIT press, 2018.

[87] P. Auer, "Using confidence bounds for exploitation-exploration tradeoffs," *Journal of Machine Learning Research*, vol. 3, no. Nov, pp. 397–422, 2002.

[88] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," in *International Conference on Learning Representations*, 2014.

[89] C. Cath, S. Wachter, B. Mittelstadt, M. Taddeo, and L. Floridi, "Governing artificial intelligence: ethical, legal and technical opportunities and challenges," *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, vol. 376, no. 2133, p. 20180080, 2018.

[90] W. Wallach and C. Allen, "Moral machines: Teaching robots right from wrong," in *Machine Ethics and Robot Ethics*. Routledge, 2008, pp. 347–363.

[91] T. Everitt, P. A. Ortega, E. Barnes, and S. Legg, "Scalable agent alignment via reward modeling: a research direction," *arXiv preprint arXiv:1811.07871*, 2018.

[92] M. Arutyunyan, A. Berestovskyy, A. Bratschi-Kaye, U. Degenbaev, M. Drijvers, I. El-Ashi, S. Kaestle, R. Kashitsyn, M. Kot, Y.-A. Pignolet, R. Rumenov, D. Sarlis, A. Sinpalean, A. Uta, B. Warinschi, and A. Zapuc, "Decentralized and stateful serverless computing on the internet computer blockchain," in *2023 USENIX Annual Technical Conference (USENIX ATC 23)*. USENIX Association, 2023, pp. 329–343.

[93] J. D. Harris and B. Waggoner, "Decentralized & collaborative ai on blockchain," *arXiv preprint arXiv:1907.07247*, 2019.